Random Forest with tuning

Libraries

```
library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.
0.0 ──
## ✓ dplyr     1.2.1     ✓ readr     2.2.0
## ✓ forcats   1.0.1     ✓ stringr   1.6.0
## ✓ ggplot2   4.0.3     ✓ tibble    3.3.1
## ✓ lubridate 1.9.5     ✓ tidyr     1.3.2
## ✓ purrr     1.2.2
## ── Conflicts ────────────────────────────────────────── tidyverse_conflict
s() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
 conflicts to become errors
```

```
library(tidymodels)
```

```
## ── Attaching packages ─────────────────────────────────── tidymodels 1.
5.0 ──
## ✓ broom        1.0.12     ✓ rsample      1.3.2
## ✓ dials        1.4.3      ✓ tailor       0.1.0
## ✓ infer        1.1.0      ✓ tune         2.1.0
## ✓ modeldata    1.5.1      ✓ workflows    1.3.0
## ✓ parsnip      1.6.0      ✓ workflowsets 1.1.1
## ✓ recipes      1.3.2      ✓ yardstick    1.4.0
## ── Conflicts ────────────────────────────────────── tidymodels_conflict
s() ──
## ✖ scales::discard() masks purrr::discard()
## ✖ dplyr::filter()   masks stats::filter()
## ✖ recipes::fixed()  masks stringr::fixed()
## ✖ dplyr::lag()      masks stats::lag()
## ✖ yardstick::spec() masks readr::spec()
## ✖ recipes::step()   masks stats::step()
```

```
library(mice)
```

```
##
## Attaching package: 'mice'
##
## The following object is masked from 'package:stats':
##
##     filter
##
```

```
## The following objects are masked from 'package:base':
##
##     cbind, rbind
```

```
library(VIM)
```

```
## Loading required package: colorspace
## Loading required package: grid
## Registered S3 method overwritten by 'car':
##   method           from
##   na.action.merMod lme4
## VIM is ready to use.
##
## Suggestions and bug-reports can be submitted at: https://github.com/statis
tikat/VIM/issues
##
## Attaching package: 'VIM'
##
## The following object is masked from 'package:recipes':
##
##     prepare
##
## The following object is masked from 'package:datasets':
##
##     sleep
```

```
library(ranger)
library(randomForest)
```

```
## randomForest 4.7-1.2
## Type rfNews() to see new features/changes/bug fixes.
##
## Attaching package: 'randomForest'
##
## The following object is masked from 'package:ranger':
##
##     importance
##
## The following object is masked from 'package:dplyr':
##
##     combine
##
## The following object is masked from 'package:ggplot2':
##
##     margin
```

```
library(caret)
```

```
## Loading required package: lattice
##
## Attaching package: 'caret'
```

```
##
## The following objects are masked from 'package:yardstick':
##
##     precision, recall, sensitivity, specificity
##
## The following object is masked from 'package:rsample':
##
##     calibration
##
## The following object is masked from 'package:purrr':
##
##     lift
```

```
library(skimr)
library(GGally)
library(gridExtra)
```

```
##
## Attaching package: 'gridExtra'
##
## The following object is masked from 'package:randomForest':
##
##     combine
##
## The following object is masked from 'package:dplyr':
##
##     combine
```

```
library(vip)
```

```
##
## Attaching package: 'vip'
##
## The following object is masked from 'package:utils':
##
##     vi
```

Load dataset

```
train <- read_csv("C:/Users/miche/OneDrive/Desktop/BAN 502 Predictive Analytics/Course Project/train.csv")
```

```
## Rows: 26570 Columns: 26
## ── Column specification ────────────────────────────────────────────────────
## Delimiter: ","
## chr  (4): product_code, attribute_0, attribute_1, failure
## dbl (22): id, loading, attribute_2, attribute_3, measurement_0, measurement_...
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
```

```
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this m
essage.
```

Structure and summary

```
str(train)
```

```
## spc_tbl_ [26,570 × 26] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
##  $ id            : num [1:26570] 0 1 2 3 4 5 6 7 8 9 ...
##  $ product_code  : chr [1:26570] "A" "A" "A" "A" ...
##  $ loading       : num [1:26570] 80.1 84.9 82.4 101.1 188.1 ...
##  $ attribute_0   : chr [1:26570] "material_7" "material_7" "material_7" "m
aterial_7" ...
##  $ attribute_1   : chr [1:26570] "material_8" "material_8" "material_8" "m
aterial_8" ...
##  $ attribute_2   : num [1:26570] 9 9 9 9 9 9 9 9 9 9 ...
##  $ attribute_3   : num [1:26570] 5 5 5 5 5 5 5 5 5 5 ...
##  $ measurement_0 : num [1:26570] 7 14 12 13 9 11 12 4 9 10 ...
##  $ measurement_1 : num [1:26570] 8 3 1 2 2 4 2 8 6 4 ...
##  $ measurement_2 : num [1:26570] 4 3 5 6 8 0 4 8 5 7 ...
##  $ measurement_3 : num [1:26570] 18 18.2 18.1 17.3 19.3 ...
##  $ measurement_4 : num [1:26570] 12.5 11.5 11.7 11.2 12.9 ...
##  $ measurement_5 : num [1:26570] 15.7 17.7 16.7 18.6 17 ...
##  $ measurement_6 : num [1:26570] 19.3 17.9 18.2 18.3 15.7 ...
##  $ measurement_7 : num [1:26570] 11.7 12.7 12.7 12.6 11.3 ...
##  $ measurement_8 : num [1:26570] 20.2 17.9 18.3 19.1 18.1 ...
##  $ measurement_9 : num [1:26570] 10.7 12.4 12.7 12.5 10.3 ...
##  $ measurement_10: num [1:26570] 15.9 17.9 15.6 16.3 17.1 ...
##  $ measurement_11: num [1:26570] 17.6 17.9 NA 18.4 19.9 ...
##  $ measurement_12: num [1:26570] 15.2 11.8 13.8 10 12.4 ...
##  $ measurement_13: num [1:26570] 15 14.7 16.7 15.2 16.2 ...
##  $ measurement_14: num [1:26570] NA 15.4 18.6 15.6 12.8 ...
##  $ measurement_15: num [1:26570] 13 14.4 14.1 16.2 13.2 ...
##  $ measurement_16: num [1:26570] 14.7 15.6 17.9 17.2 16.4 ...
##  $ measurement_17: num [1:26570] 764 682 663 826 580 ...
##  $ failure       : chr [1:26570] "No" "No" "No" "No" ...
##  - attr(*, "spec")=
##   .. cols(
##   ..   id = col_double(),
##   ..   product_code = col_character(),
##   ..   loading = col_double(),
##   ..   attribute_0 = col_character(),
##   ..   attribute_1 = col_character(),
##   ..   attribute_2 = col_double(),
##   ..   attribute_3 = col_double(),
##   ..   measurement_0 = col_double(),
##   ..   measurement_1 = col_double(),
##   ..   measurement_2 = col_double(),
##   ..   measurement_3 = col_double(),
##   ..   measurement_4 = col_double(),
```

```
##   ..   measurement_5 = col_double(),
##   ..   measurement_6 = col_double(),
##   ..   measurement_7 = col_double(),
##   ..   measurement_8 = col_double(),
##   ..   measurement_9 = col_double(),
##   ..   measurement_10 = col_double(),
##   ..   measurement_11 = col_double(),
##   ..   measurement_12 = col_double(),
##   ..   measurement_13 = col_double(),
##   ..   measurement_14 = col_double(),
##   ..   measurement_15 = col_double(),
##   ..   measurement_16 = col_double(),
##   ..   measurement_17 = col_double(),
##   ..   failure = col_character()
##   .. )
##  - attr(*, "problems")=<pointer: 0x0000022f7f6fe6f0>
```

```
summary(train)
```

```
##        id            product_code      loading           attribute_0
##  Min.   :    0   Length   :26570   Min.   : 33.16   Length   :26570
##  1st Qu.: 6642   N.unique :    5   1st Qu.: 99.99   N.unique :    2
##  Median :13284   N.blank  :    0   Median :122.39   N.blank  :    0
##  Mean   :13284   Min.nchar:    1   Mean   :127.83   Min.nchar:   10
##  3rd Qu.:19927   Max.nchar:    1   3rd Qu.:149.15   Max.nchar:   10
##  Max.   :26569                     Max.   :385.86
##                                    NAs    :250
##     attribute_1     attribute_2     attribute_3   measurement_0
##  Length   :26570   Min.   :5.000   Min.   :5.00   Min.   : 0.000
##  N.unique :    3   1st Qu.:6.000   1st Qu.:6.00   1st Qu.: 4.000
##  N.blank  :    0   Median :6.000   Median :8.00   Median : 7.000
##  Min.nchar:   10   Mean   :6.754   Mean   :7.24   Mean   : 7.416
##  Max.nchar:   10   3rd Qu.:8.000   3rd Qu.:8.00   3rd Qu.:10.000
##                    Max.   :9.000   Max.   :9.00   Max.   :29.000
##
##  measurement_1    measurement_2    measurement_3   measurement_4
##  Min.   : 0.000   Min.   : 0.000   Min.   :13.97   Min.   : 8.008
##  1st Qu.: 5.000   1st Qu.: 4.000   1st Qu.:17.12   1st Qu.:11.051
##  Median : 8.000   Median : 6.000   Median :17.79   Median :11.733
##  Mean   : 8.233   Mean   : 6.257   Mean   :17.79   Mean   :11.732
##  3rd Qu.:11.000   3rd Qu.: 8.000   3rd Qu.:18.47   3rd Qu.:12.410
##  Max.   :29.000   Max.   :24.000   Max.   :21.50   Max.   :16.484
##                                    NAs    :381     NAs    :538
##  measurement_5   measurement_6   measurement_7    measurement_8
##  Min.   :12.07   Min.   :12.71   Min.   : 7.968   Min.   :15.22
##  1st Qu.:16.44   1st Qu.:16.84   1st Qu.:11.045   1st Qu.:18.34
##  Median :17.13   Median :17.52   Median :11.712   Median :19.02
##  Mean   :17.13   Mean   :17.51   Mean   :11.717   Mean   :19.02
##  3rd Qu.:17.80   3rd Qu.:18.18   3rd Qu.:12.391   3rd Qu.:19.71
##  Max.   :21.43   Max.   :21.54   Max.   :15.419   Max.   :23.81
```

```
##  NAs    :676     NAs    :796     NAs    :937      NAs    :1048
##  measurement_9    measurement_10   measurement_11  measurement_12
##  Min.   : 7.537   Min.   : 9.323   Min.   :12.46   Min.   : 5.167
##  1st Qu.:10.757   1st Qu.:15.209   1st Qu.:18.17   1st Qu.:10.703
##  Median :11.430   Median :16.127   Median :19.21   Median :11.717
##  Mean   :11.431   Mean   :16.118   Mean   :19.17   Mean   :11.702
##  3rd Qu.:12.102   3rd Qu.:17.025   3rd Qu.:20.21   3rd Qu.:12.709
##  Max.   :15.412   Max.   :22.479   Max.   :25.64   Max.   :17.663
##  NAs    :1227     NAs    :1300     NAs    :1468    NAs    :1601
##  measurement_13  measurement_14  measurement_15   measurement_16
##  Min.   :10.89   Min.   : 9.14   Min.   : 9.104   Min.   : 9.701
##  1st Qu.:14.89   1st Qu.:15.06   1st Qu.:13.957   1st Qu.:15.268
##  Median :15.63   Median :16.04   Median :14.969   Median :16.436
##  Mean   :15.65   Mean   :16.05   Mean   :14.996   Mean   :16.461
##  3rd Qu.:16.37   3rd Qu.:17.08   3rd Qu.:16.018   3rd Qu.:17.628
##  Max.   :22.71   Max.   :22.30   Max.   :21.626   Max.   :24.094
##  NAs    :1774    NAs    :1874    NAs    :2009    NAs    :2110
##  measurement_17        failure
##  Min.   : 196.8   Length   :26570
##  1st Qu.: 619.0   N.unique :    2
##  Median : 701.0   N.blank  :    0
##  Mean   : 701.3   Min.nchar:    2
##  3rd Qu.: 784.1   Max.nchar:    3
##  Max.   :1312.8
##  NAs    :2284
```

Delete ID column

```
train = train %>% select(- id)
```

Convert character variables to factors

```
train = train %>% mutate_if(is.character, as_factor)
```

Convert other categorical variables to factors

```
train <- train %>%
  mutate(
    attribute_2 = as.factor(attribute_2),
    attribute_3 = as.factor(attribute_3)
  )
```

Check for missing data

```
skim(train)
```

Data summary

Name	train
Number of rows	26570
Number of columns	25

Column type frequency:	
factor	6
numeric	19
Group variables	None

Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
product_code	0	1	FALSE	5	C: 5765, E: 5343, B: 5250, D: 5112
attribute_0	0	1	FALSE	2	mat: 21320, mat: 5250
attribute_1	0	1	FALSE	3	mat: 10865, mat: 10362, mat: 5343
attribute_2	0	1	FALSE	4	6: 10455, 5: 5765, 8: 5250, 9: 5100
attribute_3	0	1	FALSE	4	8: 11015, 9: 5343, 6: 5112, 5: 5100
failure	0	1	FALSE	2	No: 20921, Yes: 5649

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
loading	250	0.99	127.83	39.03	33.16	99.99	122.39	149.15	385.86	
measurement_0	0	1.00	7.42	4.12	0.00	4.00	7.00	10.00	29.00	
measurement_1	0	1.00	8.23	4.20	0.00	5.00	8.00	11.00	29.00	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
measurement_2	0	1.00	6.26	3.31	0.00	4.00	6.00	8.00	24.00	
measurement_3	381	0.99	17.79	1.00	13.97	17.12	17.79	18.47	21.50	
measurement_4	538	0.98	11.73	1.00	8.01	11.05	11.73	12.41	16.48	
measurement_5	676	0.97	17.13	1.00	12.07	16.44	17.13	17.80	21.42	
measurement_6	796	0.97	17.51	1.00	12.71	16.84	17.52	18.18	21.54	
measurement_7	937	0.96	11.72	1.00	7.97	11.04	11.71	12.39	15.42	
measurement_8	1048	0.96	19.02	1.01	15.22	18.34	19.02	19.71	23.81	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
measurement_9	1227	0.95	11.43	1.00	7.54	10.76	11.43	12.10	15.41	
measurement_10	1300	0.95	16.12	1.41	9.32	15.21	16.13	17.02	22.48	
measurement_11	1468	0.94	19.17	1.52	12.46	18.17	19.21	20.21	25.64	
measurement_12	1601	0.94	11.70	1.49	5.17	10.70	11.72	12.71	17.66	
measurement_13	1774	0.93	15.65	1.16	10.89	14.89	15.63	16.37	22.71	
measurement_14	1874	0.93	16.05	1.49	9.14	15.06	16.04	17.08	22.30	
measurement_15	2009	0.92	15.00	1.55	9.10	13.96	14.97	16.02	21.63	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
measurement_16	2110	0.92	16.46	1.71	9.70	15.27	16.44	17.63	24.09	
measurement_17	2284	0.91	701.27	123.30	196.79	618.96	701.02	784.09	1312.79	

Imputation

```
set.seed(1234)
imp_train = mice(train, m=5, method="pmm", printFlag=FALSE)

## Warning: Number of logged events: 800

train_complete = complete(imp_train)
summary(train_complete)

##  product_code    loading           attribute_0        attribute_1
##  A:5100       Min.   : 33.16   material_7:21320   material_8:10865
##  B:5250       1st Qu.:100.21   material_5: 5250   material_5:10362
##  C:5765       Median :122.66                      material_6: 5343
##  D:5112       Mean   :128.01
##  E:5343       3rd Qu.:149.64
##               Max.   :385.86
##  attribute_2 attribute_3 measurement_0    measurement_1    measurement_2

##  5: 5765     5: 5100     Min.   : 0.000   Min.   : 0.000   Min.   : 0.000

##  6:10455     6: 5112     1st Qu.: 4.000   1st Qu.: 5.000   1st Qu.: 4.000

##  8: 5250     8:11015     Median : 7.000   Median : 8.000   Median : 6.000

##  9: 5100     9: 5343     Mean   : 7.416   Mean   : 8.233   Mean   : 6.257

##                          3rd Qu.:10.000   3rd Qu.:11.000   3rd Qu.: 8.000

##                          Max.   :29.000   Max.   :29.000   Max.   :24.000

##  measurement_3   measurement_4    measurement_5   measurement_6
##  Min.   :13.97   Min.   : 8.008   Min.   :12.07   Min.   :12.71
##  1st Qu.:17.12   1st Qu.:11.072   1st Qu.:16.45   1st Qu.:16.84
```

```
##  Median :17.78   Median :11.735   Median :17.11   Median :17.54
##  Mean   :17.79   Mean   :11.739   Mean   :17.12   Mean   :17.56
##  3rd Qu.:18.48   3rd Qu.:12.417   3rd Qu.:17.80   3rd Qu.:18.23
##  Max.   :21.50   Max.   :16.484   Max.   :21.43   Max.   :21.54
##  measurement_7    measurement_8   measurement_9    measurement_10
##  Min.   : 7.968   Min.   :15.22   Min.   : 7.537   Min.   : 9.323
##  1st Qu.:10.978   1st Qu.:18.35   1st Qu.:10.761   1st Qu.:15.260
##  Median :11.686   Median :19.03   Median :11.440   Median :16.213
##  Mean   :11.686   Mean   :19.02   Mean   :11.407   Mean   :16.245
##  3rd Qu.:12.384   3rd Qu.:19.70   3rd Qu.:12.093   3rd Qu.:17.197
##  Max.   :15.419   Max.   :23.81   Max.   :15.412   Max.   :22.479
##  measurement_11  measurement_12   measurement_13  measurement_14
##  Min.   :12.46   Min.   : 5.167   Min.   :10.89   Min.   : 9.14
##  1st Qu.:18.24   1st Qu.:10.659   1st Qu.:14.90   1st Qu.:15.07
##  Median :19.32   Median :11.692   Median :15.65   Median :15.95
##  Mean   :19.31   Mean   :11.625   Mean   :15.68   Mean   :16.02
##  3rd Qu.:20.41   3rd Qu.:12.756   3rd Qu.:16.41   3rd Qu.:17.06
##  Max.   :25.64   Max.   :17.663   Max.   :22.71   Max.   :22.30
##  measurement_15   measurement_16   measurement_17   failure
##  Min.   : 9.104   Min.   : 9.701   Min.   : 196.8   No :20921
##  1st Qu.:14.037   1st Qu.:15.102   1st Qu.: 628.3   Yes: 5649
##  Median :14.996   Median :16.316   Median : 715.1
##  Mean   :15.023   Mean   :16.281   Mean   : 729.4
##  3rd Qu.:15.995   3rd Qu.:17.596   3rd Qu.: 805.9
##  Max.   :21.626   Max.   :24.094   Max.   :1312.8
```

Split data

```
set.seed(123)
train_complete_split = initial_split(train_complete, prop=0.7, strata=failur
e)
train_set = training(train_complete_split)
test_set = testing(train_complete_split)
```

Set up folds for cross-validation

```
set.seed(123)
rf_folds = vfold_cv(train_set, v = 5)
```

Random forest

```
train_complete_recipe = recipe(failure ~ loading, train_set) %>%
  step_dummy(all_nominal(), -all_outcomes())

rf_model = rand_forest() %>%
  set_engine("ranger", importance = "permutation") %>%
  set_mode("classification")

train_complete_wflow =
  workflow() %>%
  add_model(rf_model) %>%
```

```
  add_recipe(train_complete_recipe)

set.seed(123)
train_complete_fit = fit(train_complete_wflow, train_set)
```

Predictions

```
trainpredrf = predict(train_complete_fit, train_set)
head(trainpredrf)
```

```
## # A tibble: 6 × 1
##   .pred_class
##   <fct>
## 1 No
## 2 No
## 3 No
## 4 Yes
## 5 No
## 6 No
```

Confusion matrix

```
confusionMatrix(trainpredrf$.pred_class, train_set$failure, positive = "Yes")
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    No   Yes
##        No  14302  2434
##        Yes   342  1520
##
##                Accuracy : 0.8507
##                  95% CI : (0.8455, 0.8558)
##     No Information Rate : 0.7874
##     P-Value [Acc > NIR] : < 2.2e-16
##
##                   Kappa : 0.4475
##
##  Mcnemar's Test P-Value : < 2.2e-16
##
##             Sensitivity : 0.38442
##             Specificity : 0.97665
##          Pos Pred Value : 0.81633
##          Neg Pred Value : 0.85457
##              Prevalence : 0.21260
##          Detection Rate : 0.08173
##    Detection Prevalence : 0.10012
##       Balanced Accuracy : 0.68053
##
##        'Positive' Class : Yes
##
```

Predictions on test set

```
testpredrf = predict(train_complete_fit, test_set)
head(testpredrf)
```

```
## # A tibble: 6 × 1
##   .pred_class
##   <fct>
## 1 No
## 2 Yes
## 3 Yes
## 4 No
## 5 Yes
## 6 No
```

```
confusionMatrix(testpredrf$.pred_class, test_set$failure, positive = "Yes")
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction   No  Yes
##        No  5574 1470
##        Yes  703  225
##
##                Accuracy : 0.7274
##                  95% CI : (0.7175, 0.7372)
##     No Information Rate : 0.7874
##     P-Value [Acc > NIR] : 1
##
##                   Kappa : 0.0249
##
##  Mcnemar's Test P-Value : <2e-16
##
##             Sensitivity : 0.13274
##             Specificity : 0.88800
##          Pos Pred Value : 0.24246
##          Neg Pred Value : 0.79131
##              Prevalence : 0.21262
##          Detection Rate : 0.02822
##    Detection Prevalence : 0.11641
##       Balanced Accuracy : 0.51037
##
##        'Positive' Class : Yes
##
```

Save the file

```
saveRDS(train_complete_fit, "train_complete_fit.rds")
```

Load the model

```
train_complete_fit = readRDS("train_complete_fit.rds")
```

Variable importance

```
train_complete_fit %>% extract_fit_parsnip() %>% vip(geom = "point")
```

